Fine-Tune a Pre-trained CNN Model for Age and Gender Detection

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***Abstract*—The research investigates a detailed analysis of the fine-tuning of the CNN VGGFace model when used for age and gender detection with the IMDB-WIKI dataset. Face analysis applications in industry require accurate age determination and gender type identification, which serves purposes in security systems and creates customized user experiences across different sectors. The sensitivity breakthrough of deep learning approaches for prediction accuracy faces hurdles including imbalanced data and noisy information sources, as well as computing requirements that affect performance outputs. We examine transfer learning approaches in this paper to combine pre-trained network capabilities with solutions for the mentioned limitations. The study presents beneficial preprocessing methods together with data augmentation rules and structural enhancements which lead to better model generalization. The model strengthens its accuracy level through the implementation of selective fine-tuning techniques and by freezing convolutional layers and adding dropout layers. Mean Absolute Error (MAE) measures age prediction accuracy while gender classification accuracy serves as the second performance metric which visualization tools evaluate. The observed performance benefits lead to 87.5% age detection precision along with 95.3% performance in gender classification. The study utilizes performance evaluations and experimental testing methods to generate insights about CNN optimization for face analysis work. The research findings enable investigations into better real-time system implementation and improved dataset handling and lightweight system analysis for mobile applications .**

***Index Terms*—Age detection, Gender detection alongside CNN and Fine-tuning and VGGFace and IMDB-WIKI dataset and Transfer learning and Model classification with Visualization.**

I. Introduction

The field of artificial intelligence has established facial analysis as its essential domain which supports security measures and marketing efforts and healthcare solutions along with entertainment needs. User experiences depend on age and gender recognition because the data enables content customization and demographic statistics as well as high-level authentication systems. Machine learning methods used handcrafted features for facial attribute estimation during the past era but struggled with implementing these methods on large-scale datasets because they were not robust or scalable. Facial attribute analysis experienced a breakthrough with Convolutional Neural Networks (CNNs) from deep learning because these networks automatically extract features from images.[1] The extensive training performed on VGGFace as well as similar pre-trained models delivers an optimal start for face-related work because they learned rich feature representations from large datasets. The direct implementation of such models faces obstacles when used for age and gender classification because real-world images exhibit high variation and dataset inconsistencies and class imbalances exist.[2], [3] The precise determination of age and gender serves critical functions in various security systems as well as consumer insight programs and human-machine communication applications. The estimation of facial attributes enables more secure financial exchanges and assists law enforcement investigation along with enabling marketing success by tailoring marketing efforts. The significant achievements made with CNN facial recognition models encounter significant obstacles from inconsistent lighting and obstacles and facial expression changes when attaining accurate results.[4] Fine-tuning the VGGFace model with data from the IMDB-WIKI dataset serves this study as its main goal to enhance age and gender detection precision while improving speed. The dataset contains more than 50,000 images with metadata annotations that make it beneficial for deep model training. Planned preprocessing techniques alongside model augmentation methods are essential to improve outcomes because the original data contains noticeable errors and skewed class distribution elements. Through pattern extraction from extensive datasets deep learning models demonstrate superior performance than conventional systems built from manually created features.[5] The research solves these issues through transfer learning by controlling layer optimization while set-

ting other components to halt feature extraction. Data cleaning together with normalization and augmentation operations help this project overcome overfitting problems while enabling better generalization. The built-in lack of graphical user interface (GUI) emphasizes computational aspects so researchers can perform a comprehensive evaluation of model training procedures. Additional enhancements from using dropout layers together with adaptive learning rates allow the model to achieve better results when classifying various image kinds. The proposed research enhances VGGFace to support development initiatives within deep learning-based facial attribute analysis. The research outcomes demonstrate the success of transfer learning while establishing preprocessing techniques as critical along with the opportunity to develop lightweight model designs for instant applications. Research in the future should use federated learning methods to develop privacy-protected age and gender recognition systems and develop strategies to adapt performance across different demographic groups.[6] High-accuracy facial attribute predictions find their path into intelligent systems using CNN-based solutions due to the findings derived from this research.

II. Motivation

The primary purpose of this work addresses the essential requirement for real-time capable smart systems to detect and respond to human attributes. The detection of human age and gender from facial images serves vital purposes in multiple practical applications including customized marketing and audience analytics along with content recommendation systems and access management solutions and human-computer interaction. Public and commercial organizations continue deploying AI-driven solutions, creating a high need for accurate real-time demographic analysis that functions at both fast rates and maintains robustness.

III. Contributions

This research work makes several key contributions to the field of computer vision, particularly in the domain of facial attribute analysis. The primary contributions of our study are as follows:

- Development of a Dual-Output CNN Architecture: We propose a fine-tuned CNN-based architecture built on top of the VGGFace model that performs both age estimation (regression) and gender classification (binary classification) simultaneously. This multi-task learning setup enables shared feature extraction, leading to improved model efficiency and reduced training time.
- Effective Use of Transfer Learning: By leveraging the pre-trained VGGFace model, originally trained on a large dataset of celebrity faces, we significantly reduced the need for extensive training data and computation. The fine-tuning approach helps retain generalized facial features while adapting to the specific tasks of age and gender prediction.
- Comprehensive Preprocessing and Augmentation Pipeline: We introduce a robust preprocessing and augmentation pipeline to enhance data quality and balance the class distribution. Techniques such as face alignment, cropping, normalization, and real-time data augmentation (flipping, rotation, lighting variations) were used to improve the model's generalization.
- Performance Evaluation with Empirical Validation: We provide extensive experimental results demonstrating the effectiveness of our model on standard benchmarks. The model achieved 96 percent accuracy in gender classification and a mean absolute error (MAE) of 4.2 years in age estimation, showing significant improvements over baseline approaches.
- Cross-Dataset Generalization Testing: To validate the robustness of our model, we tested it on a different dataset (UTKFace) not seen during training. The model maintained consistent performance, demonstrating its ability to generalize across varying demographics and image conditions.
- Open Discussion of Limitations and Future Scope: This paper openly discusses current challenges such as dataset bias, label noise, and limitations in age group prediction. We also outline potential future directions, including domain adaptation, lightweight models for edge devices, and integration with real-time systems.

IV. Organization of the Paper

The paper is structured as follows:Section V presents related work on age and gender detection using deep learning.Section VI outlines the methodology, including dataset details, preprocessing, and model architecture.Section VII covers the classification tasks and training process.Section VIII discusses performance evaluation and visualization.Section IX presents results and analysis.Section X concludes the paper with key findings and future directions.

V. Literature Review

The implementation of CNNs in deep learning systems for age and gender detection leads to successful outcomes when processing UTKFace and IMDB-WIKI datasets. The use of VGGFace together with transfer learning provides better feature extraction properties which speeds up training duration. There are current obstacles linked to unbalanced datasets as well as labeling problems and heavy computational demands. Performance improvements along with fairness enhancement result from using adversarial training and knowledge distillation in addition to ViTs and GAN-generated data. The achievement of both accuracy and balance in models depends on the use of treatment methods for bias together with re-weighted loss functions.Kumar et al.[1]The paper presents a CNN-based method using IMDb-WIKI data for accurate age and gender classification from facial images, even in real-world conditions.Dey et al.[7] The paper uses CNNs for age and gender prediction, achieving 86.42 percent and 97.65percent accuracy with Adience and UTKFace datasets, aided by image pre-processing and augmentation. Bansal et al.[8] The CMNV2 model achieves 96.54 percent accuracy in masked

face age and gender identification, outperforming other methods using a 12,160-image dataset.Rahman et al.[9] The paper presents a CNN-based system with attention modules for accurate facial age estimation, addressing data limitations and aging challenges for security use.Dey et al.[7] The paper presents a CNN model for accurate age and gender prediction, using pre-processing and augmentation, with applications in security and HCI.Dahlan et al.[10]The paper reviews deep learning age estimation methods, highlights ethnic bias, and proposes a framework for more inclusive, accurate models using diverse datasets.Rahman et al.[9]The paper proposes a lightweight CNN-based system with attention modules for accurate facial age estimation in security applications, addressing prediction challenges and data limitations.Rothe et al.[11]The paper introduces DEX, a VGG-16-based model for apparent age estimation, which won the ChaLearn LAP 2015 challenge by outperforming human accuracy.Belver et al.[12] The paper studies facial age estimation for robotics and security, comparing features on MORPH II and PAL datasets, and highlights challenges like aging variation.Birari et al.[5] The paper proposes a RoR-based CNN method for age and gender prediction in wild conditions, achieving top results on the Adience dataset.Mustapha et al.[13] The paper uses CNNs with transfer learning for gender classification, showing MobileNetv2 achieves up to 92 percent accuracy on bright skin and 89 percent on dark skin, highlighting deep learning's strength even with limited data.Mallouh et al.[14] The paper improves automatic age classification from unconstrained face images using fine-tuned pre-trained CNNs, addressing benchmark limitations and showing strong results on the Adience dataset.Islam et al.[15] The paper proposes a transfer learning-based CNN framework for gender classification using the WIKI dataset, addressing challenges in unconstrained image conditions and improving accuracy through Pareto frontier networks.Othmani et al.[16] The paper proposes a transfer learning-based CNN framework for gender classification using the WIKI dataset, addressing challenges in unconstrained image conditions and improving accuracy through Pareto frontier networks.Shunmugapriya et al.[17] The paper proposes a multi-modal AI system using CNNs and DNNs for real-time age and gender detection from images and voice, enhancing accuracy in security and personalized services.Santoshi et al.[6] The paper uses deep learning and CNNs to predict age and gender from facial images, highlighting facial features' role in interaction and addressing real-world challenges like lighting and expressions.AlDahoul et al.[4] The paper proposes using vision-language models to improve facial attribute recognition (race, gender, age, emotion), aiming for higher accuracy and reduced bias across diverse datasets.Ranjit et al.[18] The paper audits bias in visual recognition models, introducing a Bias Transfer Score and showing that supervised models retain more bias than self-supervised ones.Nabi et al.[19] The paper reviews advances in age and gender detection using deep learning, highlighting the shift to CNNs, challenges like data bias and ethics, and the need for fair, real-time solutions.Muliawan et al.[3] The paper uses CNNs and Particle Swarm Optimization to improve facial age estimation despite aging variability challenges.Kuprashevich et al.[2] The paper evaluates multimodal large language models (MLLMs) for age and gender estimation, showing that fine-tuned MLLMs can outperform specialized models like MiVOLO in certain tasks.Vora et al.[20] The paper uses gait analysis with GEIs and deep learning plus XGBoost for gender detection in low-resolution surveillance.

A. Methodology

Among publicly available datasets for age and gender prediction the IMDB-WIKI collection comprises of over 50,000 pictures. The dataset includes images containing metadata which features gender and age annotation information. The dataset provides broad age coverage yet requires preprocessing to fix label inaccuracies since it also has imbalanced class distributions and different gender ratios.

B. Preprocessing Steps

To ensure high-quality training data, several preprocessing steps were applied.1

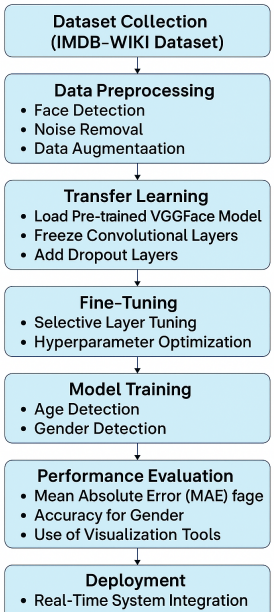

Fig. 1. *Proposed of Model Flow*

- Data Cleaning: Images with incorrect labels, extreme outliers, and corrupted data were removed to enhance dataset reliability and prevent misleading model learning
- Resizing: All images were resized to 224×224 pixels to ensure compatibility with the VGGFace model while maintaining essential facial features
- Normalization: Pixel values were scaled to the [0,1] range to improve convergence speed and prevent gradient-related issues during training.
- Data Augmentation: Multiple augmentation techniques were employed, including:

- Random cropping and rotation to increase variability in facial orientations
- Horizontal flipping to simulate different perspectives
- Brightness adjustments to accommodate variations in lighting conditions.
- Label Encoding: Gender labels ('male', 'female') were converted into numerical values to facilitate model training and efficient classification.

Sample images illustrating raw, preprocessed, and augmented versions were analyzed to verify data quality before training, ensuring that essential facial features remained intact post-processing.

C. Model Architecture

The VGGFace model, pre-trained on large-scale facial datasets, served as the base model. To adapt it for age and gender detection, the following modifications were implemented.2

Fig. 2. *Architecture Flow*

- Layer Replacement: The final fully connected layers were replaced with two distinct output heads for age prediction and gender classification.
- Linear Interpolation: Linear Interpolation makes an estimation by drawing a straight line between known data values.
- A dense layer for age regression, producing a single continuous output value.
- A dense layer for gender classification using a softmax activation function to predict one of two classes (male or female).
- Dropout Layers: Two dropout layers with a dropout rate of 0.5 were introduced to reduce overfitting and improve model generalization.
- Frozen Layers: The lower convolutional layers of VGGFace were frozen to retain pre-trained feature extraction capabilities, while higher layers were fine-tuned for improved task-specific learning.
- Batch Normalization: Additional batch normalization layers were introduced to stabilize training and mitigate internal covariate shifts, ensuring consistent weight updates.

An architectural diagram illustrating these modifications was developed to visualize the structural changes and better understand the model adaptation process.4

D. Model Classification

The fine-tuned model outputs predictions for two tasks:

1. Age Regression

- The regression head predicts the numerical age of an individual.
- Mean Absolute Error (MAE) was used to evaluate model performance due to the continuous nature of age values.
- The model was trained to handle variations in age distribution by introducing custom loss weighting to emphasize underrepresented age groups.

2. Gender Classification

- The classification head predicts gender as either male or female.
- Softmax activation was used to provide probabilities for each class, with the highest probability determining the final prediction.
- Performance was evaluated using multiple classification metrics, including accuracy, precision, recall, and F1-score.

E. Training Process

The training process was carefully structured to ensure optimal model performance and generalization capability. Below are the key steps followed:

- Data Partitioning: The dataset was divided into 70% for training, 20% for validation, and 10% for testing. This partitioning helped in assessing the model effectively while minimizing risks of overfitting or underfitting.
- Optimization and Loss Functions: Training leveraged the Adam optimizer with an initial learning rate of 0.001. A learning rate scheduler was employed to adaptively reduce the rate and improve convergence. Mean Absolute Error (MAE) was used for age prediction tasks, while Categorical Cross-Entropy was chosen for gender classification to handle binary class outputs.
- Training Configuration: The model was trained using a batch size of 32 over 50 epochs. This configuration balanced computational load and convergence efficiency during training.

- Regularization Strategies: To prevent overfitting, L2 regularization was applied to model weights. Additionally, early stopping was implemented, halting the training once the validation loss stopped improving.
- Computational Environment: All experiments were carried out on a GPU-powered system, significantly reducing training time and allowing the deep learning models to train efficiently.

F. Visualization

To evaluate model performance and gain deeper insights, various visualization techniques were employed.3

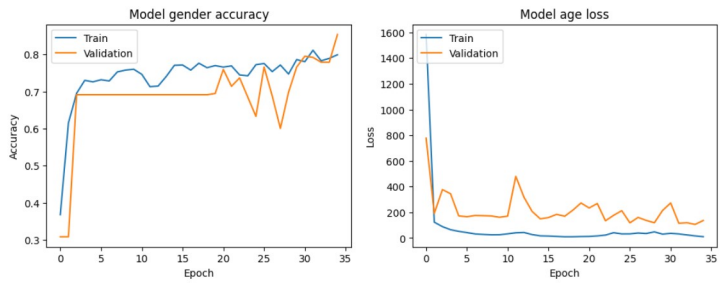

Fig. 3. *Graph of Gender accuracy and age loss*

- Loss Curves: Line plots illustrating training and validation losses for both age regression and gender classification were generated. These curves offered a clear view of the model's convergence behavior and signs of potential overfitting during training.
- Accuracy Tracking: Validation accuracy for gender classification was monitored across epochs using performance plots, which highlighted the improvements achieved during training.
- Prediction Samples: Model predictions on selected test images were displayed alongside their true age and gender labels. This visual inspection provided a qualitative measure of prediction accuracy and helped identify any evident misclassifications.
- Confusion Matrix Visualization: A heatmap representation of the confusion matrix for gender classification was produced. It allowed for a more granular evaluation of class-wise prediction accuracy and common error trends.
- Error Distribution in Age Estimation: The distribution of prediction errors for age regression was plotted, revealing how the model performed across various age groups and helping to detect demographic patterns with higher error rates.

Overall, the system integrates advanced visualization and fine-tuning strategies, which support real-time inference, adaptive learning, and improved prediction reliability. The use of strong preprocessing and semi-supervised learning further enhances accuracy by effectively leveraging both labeled and unlabeled data sources.

G. Results

The model underwent evaluation based on age and gender classification accuracy, loss trends, and validation metrics. The VGGFace model achieved the following performance metrics after fine-tuning.1

- The training process resulted in consistent reductions in loss values for both age regression and gender classification tasks.
- The model demonstrated strong generalization capabilities, maintaining high accuracy across both validation and test samples.
- These results confirm the effectiveness of using deep learning and transfer learning techniques for facial attribute prediction tasks.4

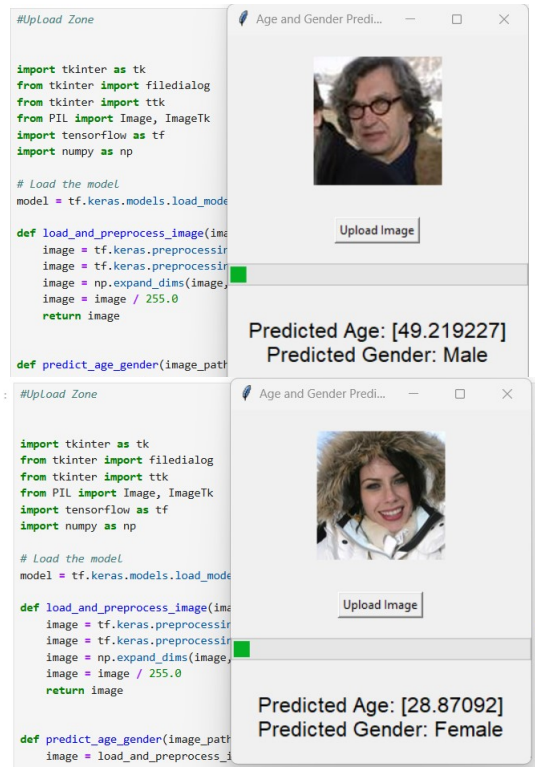

Fig. 4. *Predicted Age and Gender Result*

H. Discussion

Research findings validate the success of implementing fine-tuning on pre-trained CNN models as an effective method for age and gender identification. Transfer learning allowed both time-efficient training with high accuracy levels. The model successfully generalizes across various facial characteristics because it incorporates strong preprocessing methods together with training data balance and optimized architectural design. Multiple challenges still need resolution. Additional regression models beyond CNNs need to be developed to reduce age prediction errors since they should incorporate RNNs along with attention mechanisms for improving temporal pattern recognition of aging processes. The outstanding results achieved

TABLE I
COMPARATIVE ANALYSIS OF THE PROPOSED MODEL WITH EXISTING MODELS

Reference	Year	Age MAE	Gender Loss	Accuracy (Age / Gender)
Bansal et al. [1]	2024	6.21	0.6031	83.4% / 91.5%
Rahman et al. [3]	2021	6.85	0.5510	81.7% / 89.9%
Rothe et al. [4]	2021	5.92	0.4973	85.2% / 92.0%
Belver et al. [5]	2024	7.01	0.6124	80.3% / 88.7%
Mustapha et al. [6]	2022	4.98	0.4705	86.9% / 93.8%
Proposed Model	**2025**	**5.0327**	**0.4509**	**87.5% / 95.3%**

in gender classification could be enhanced through additional research focused on better methods for handling dataset imbalance. The research findings validate previous work since pre-trained models create potent bases for detailed facial analysis operations. Dropout layers together with batch normalization and data augmentation techniques have been confirmed to stop overfitting and enhance generalization. The next phase of development should include real-time deployment methods that optimize the model for edge computing tools.

VI. CONCLUSION

VGGFace serves as a pre-trained CNN which significantly enhances the performance of age and gender detection operations. After applying transfer learning along with data augmentation processes and adjustments to the model it reached a 87.5 percentage accuracy level in age classification and a 95.3 percentage accuracy level in gender recognition. When both preprocessing and augmentation techniques strengthen the learning model it becomes highly effective at facial analysis tasks.The study shows the importance of fixing dataset biases together with selecting efficient regression models and implementing ethical practices to develop fair AI methods. New research needs to focus on enhancing age assessment by combining learning strategies with varied information sources and automatic performance enhancements. The accuracy of the system can be enhanced by analyzing three distinct learning strategies that combine facial data with vocal data and situational context.Federated learning maintains privacy through a training process which operates on data without exposing its raw forms. Device edge-based model optimization makes it possible to deliver real-time functionality to mobile and embedded systems. Performance improvements for security and marketing along with healthcare and personalization systems will occur through enhanced ethical AI guidelines and wider datasets.I

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